

Rafael Diaz

 [rafael-diaz1](#)  [availe](#)  [availe.io](#)

Junior Software Engineer with experience in web development and the Kotlin/JVM ecosystem.

Education

College of Staten Island
B.S. in Computer Science

Graduated: Dec 2024

Technical Skills

Languages: Kotlin (JVM & Multiplatform), Python, C++, JavaScript/TypeScript, HTML/CSS

Technologies: Spring Boot, Compose Multiplatform, Gradle, Next.js, Angular

Experience

Software Engineering Intern | Newmark, New York, NY

Jun 2024 – Aug 2024

- Developed Angular UI components with routing for managing and previewing web crawler results.
- Migrated large sections of global SCSS to component-scoped styles to reduce style conflicts and improve maintainability.
- Built an Angular page selector that handled progressively updated page count data from a legacy backend; used lifecycle hooks to reduce UI delay during re-renders.
- Received positive feedback and a referral from the project's Tech Lead.

Linux Technician Intern | Innovare, Remote (Peru)

Sep 2023 – Nov 2023

- Migrated servers to Sophos Central for endpoint security, cutting infrastructure costs and simplifying management.
- Configured hostnames, IP ranges, repository files, and user accounts for new server installs.
- Wrote a C++ tool to automate generation of config scripts and batch files.

WordPress and Web Design Intern | Blackfacts, Remote (USA)

Jul 2023 – Aug 2023

- Helped build the Blackfacts for Schools landing page with WordPress and Elementor.
- Built UI components with Elementor and designed layout concepts in Figma.

Projects

KReplica | Kotlin, Gradle, KSP, Kotlin Poet

(Website) (GitHub)

- Built a compile-time code generation tool using KotlinPoet to define multiple CRUD DTO variants (data, patch, create) from a single interface. Produces plain Kotlin files for clarity and maintainability.
- Utilized the Kotlin Symbol Processing (KSP) API to enable defining the KReplica interface via nested annotations for seamless IDE autocomplete.
- Leveraged sealed interfaces to enable exhaustive `when` filtering for DTOs by variant or version.
- Published as a zero-config Gradle plugin on Maven Central.

KReplica Docs/Code Playground | Spring Boot, Kotlin, JTE, JavaScript, HTMX, CSS

(Website) (GitHub)

- Built a documentation website for KReplica with Spring Boot, JTE, and HTMX.
- Implemented a code playground where users can test out KReplica code in real-time.
- Used Caffeine (Java) to cache codegen output, with a permanent cache for documentation examples and temporary cache for user-submitted input.
- Avoided repetitive user-compilation requests by using normalized code input as cache key. Added a per-tab UUID identifier to avoid spam without needing to use IP addresses or user accounts as identifiers.
- Deployed the website as containerized application on a VPS via Podman/Shell scripts.

Other Projects

(Website)

- Frontend: Build fancy zoomable charts with shadcn/ui ([website](#))
- Resume Builder: Built with NextJS and SwiftLatex. Features draggable UI. ([website](#))
- Assembly: Built a physical counter with a PIC18F452 and two 7-segment displays ([video](#))